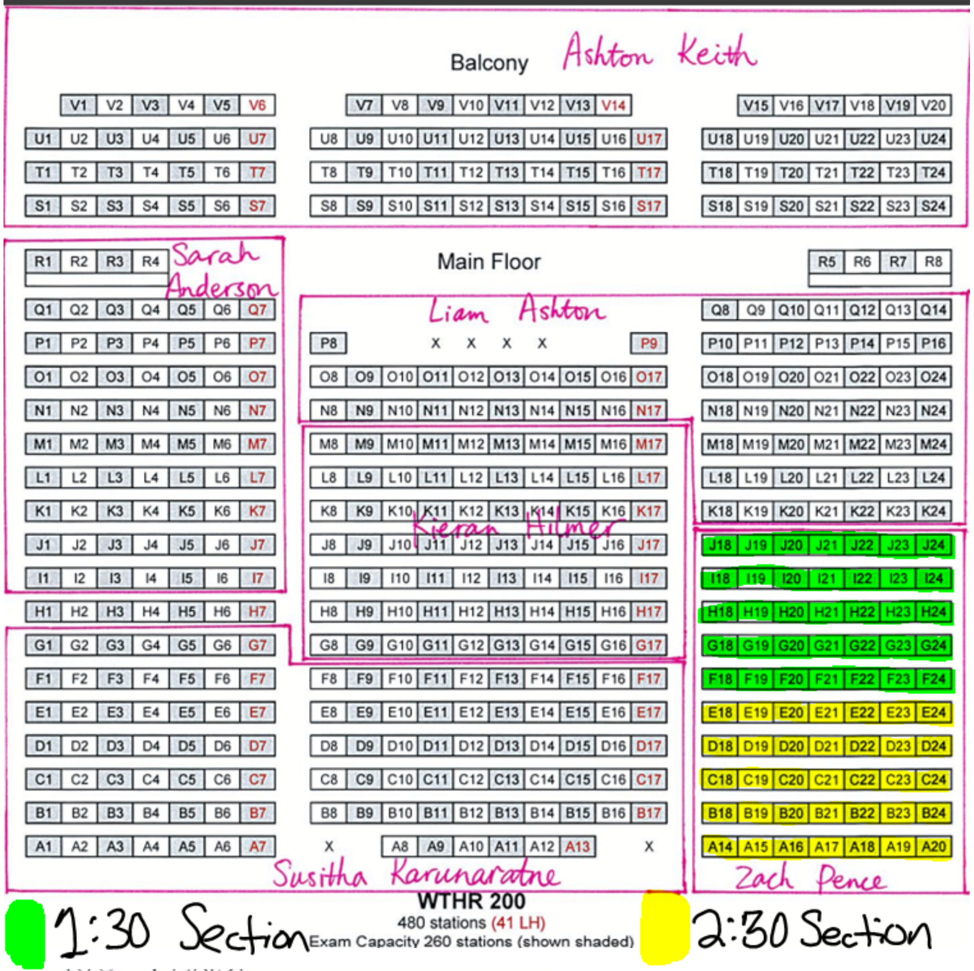
MT 2 Review

- Tuesday (11/10)
- 8-9 PM ← Show up at 7:45 - 7:50
- Lectures 19-28 [See Below]
- WTHR 200 ← Main Floor NOT Balcony

Additional Office Hours this week

- Tuesday, 12:30-2:30 in my office (MATH 615)
- Usual office hours (Tues. 2:30-3:00, MATH G175)



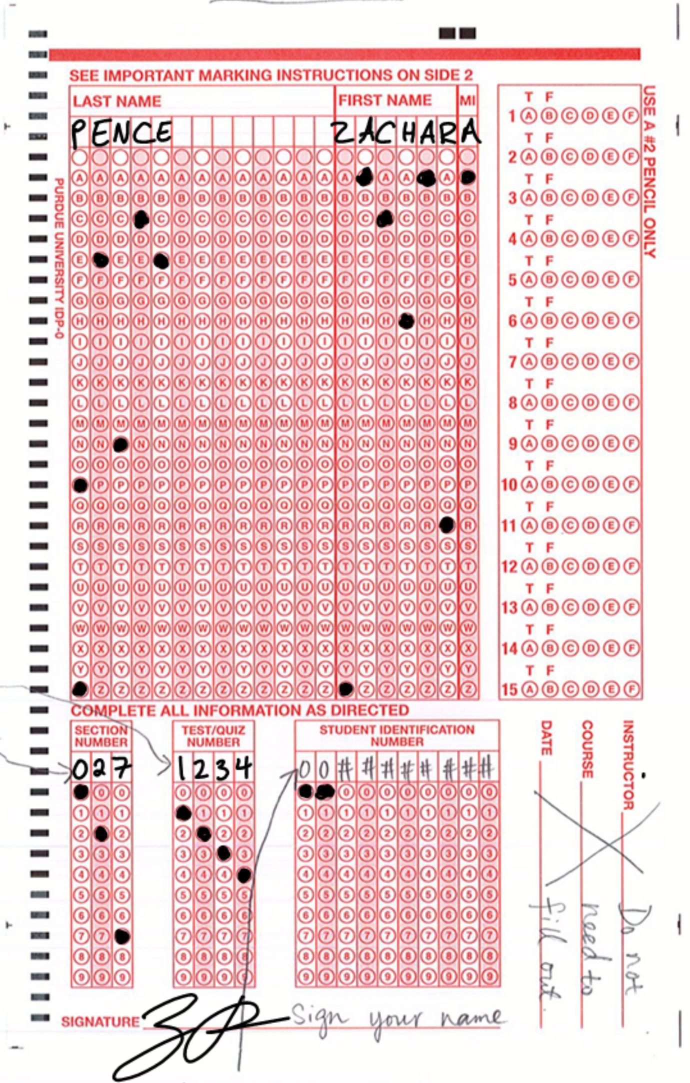
Balcony
Ashton Keith
Main Floor
Sarah Anderson
Liam Ashton
Kieran Hilmer
Susitha Karunaratne
Zach Pence
WTHR 200
480 stations (41 LH)
Exam Capacity 260 stations (shown shaded)

🟩 1:30 Section

🟨 2:30 Section

Make Sure to Bring:

- Pencil / Erasers
- PUID
- Calculator TI 30XA Single Line Only



SEE IMPORTANT MARKING INSTRUCTIONS ON SIDE 2

LAST NAME	FIRST NAME	MI
PENCE	ZACHARA	

USE A #2 PENCIL ONLY

PURDUE UNIVERSITY IDP-0

COMPLETE ALL INFORMATION AS DIRECTED

SECTION NUMBER	TEST/QUIZ NUMBER	STUDENT IDENTIFICATION NUMBER
027	1234	00########

on the exam cover page

Should start with two zeros

DATE / COURSE / INSTRUCTOR — Do not need to fill out.

SIGNATURE — Sign your name

NOTE: These are just review problems of Lectures 19-28; these are not necessarily representative of the problems of the exam. The exam can (and most likely will) have different problems.

Topics of Exam 3:

- **Lecture 19:** Absolute Extrema on an Interval
- **Lecture 20:** Graphical Interpretation of Derivatives
- **Lecture 21:** Limits at Infinity
- **Lecture 22:** A Summary of Curve Sketching
- **Lecture 23-25:** Optimization
- **Lecture 26-27:** Antiderivatives and Indefinite Integration
- **Lecture 28:** Area and Riemann Sums

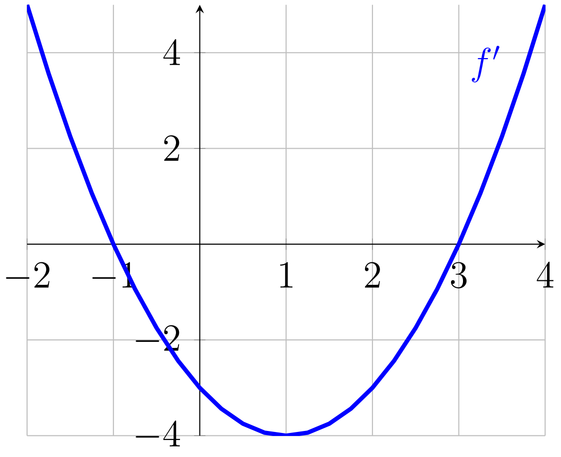
Problem 1. *Let M and m be the maximum and minimum of $f(x) = x^3 - 3x^2 + 3$ on the interval $[1, 4]$, respectively. What is the value of $M + m$?*

Problem 2. *A graph of f' is given below.*

(a) Determine when f is increasing and when it is decreasing.

(b) Determine when f is concave up and when it is concave down.

(c) Locate the positions (x-coordinates) of any relative extrema and inflection points.



Problem 3. *Compute* $\lim_{x\to\infty} \frac{x^2+1}{2-x}$ *and* $\lim_{x\to-\infty} \frac{x^2+1}{2-x}$.

Problem 4. *Use the techniques learned in Lecture 22 to sketch a graph of the function* $y = \frac{x^2}{x+8}$.

Problem 5. *If a rectangle has a fixed perimeter of* 40, *what is its maximum area?*

Problem 6. *Find the point on the line* $6x + y = 9$ *that is closest to the point* $(-3, 1)$.

Problem 7. *A particle is moving in a straight line; its acceleration function is given by* $a(t) = \cos t + \sin t$ *for* $t \geq 0$*. Find the position of the particle at time* t *when the particle has an initial position of 0 and an initial velocity of 5.*

Problem 8. *We estimate the area underneath the graph of* $f(x) = \frac{1}{x}$ *on the interval* $[1, 5]$.

- (a) *Compute the left Riemann sum with* 4 *rectangles. Is this an overestimate or an underestimate? Explain why (The exact area is* $\ln 5$*, but you don't need to know that to answer the question).*
- (b) *Repeat (a) using the right Riemann sum.*
- (c) *Set up (but do not compute) the left Riemann sum using* N *rectangles (where* N *is a positive integer).*